



Department: Information Technology

Bachelor of Information and Communications Technology (BINCTI) & Advanced Diploma in Information and Communications Technology (ADICTI)

MODULE / SUBJECT CODE: PBDV30I & PBDE40I

Project I – Web-Based Application Project

Due Date – 23 March 2026

Marks - Refer to the Assessment Rubric

General Information and Project Task

Collaborative projects afford students the opportunity to fully engage in their learning by communicating their needs, ideas, and opinions. This not only helps to cultivate competencies in various areas of software and application development, but also prepares them for future employment opportunities and other projects.

Establishment of the Project Team

This is a group project evaluation. The project teams, consisting of 10–13 members per group, were established by the lecturer based on the responses gathered from the survey administered to students. The survey results were used to ensure balanced representation of diversity and inclusiveness in terms of backgrounds, perspectives, and capabilities within each group.

Each group is required to provide the following information:

1. **Group Name:** State a unique name that best represents your group.
2. **Group Leader:** Indicate the elected group leader who will coordinate team activities and communication.

Each group member must:

- Contribute to overall project objectives.
- Complete individual deliverables.
- Provide expertise.
- Work respectfully with other group members.
- Find solutions to problems and not make excuses.

Note that a member of one group may not be allowed to migrate to another group. Any unforeseen challenges should be discussed with the lecturer. While this project emphasizes group work, it is the responsibility of each individual student to ensure that they understand the curriculum for this module and the requirements of the project.

Project Themes

This presents the selected project topics, the challenges they aim to address, and the proposed web-based solutions for implementation. Each student group has been assigned a project designed to tackle real-world campus and community issues. Some projects focus on improving campus operations and student experiences at DUT, while others address broader societal challenges, providing students with the opportunity to develop impactful, user-oriented digital solutions.

The “**Need for Solution**” provided under each topic serves as a guiding direction rather than a complete specification. It outlines the expected focus and core objectives of the project. Students are expected to critically analyse the problem, expand on the suggested direction, and propose additional innovative features and improvements to develop a comprehensive and well-justified web-based solution

Group 24 & 2 – Digital Grievance & Complaint Portal

- **Problem:** DUT Students often submit complaints manually or via scattered channels, leading to delayed resolution and poor tracking.
- **Need for Solution:** Develop a web-based solution to submit, track, and resolve grievances efficiently, ensuring accountability and timely responses.

Group 20 & 4 – Res/Room Maintenance Request System

- **Problem:** Maintenance issues in hostels are reported manually, often lost or delayed.
- **Need for Solution:** Build a web-based solution to submit, track, and manage repair requests with admin oversight, ensuring faster resolution and proper record-keeping.

Group 32 & 6 – Alumni Directory & Communication Portal

- **Problem:** Alumni data is fragmented, limiting mentorship and networking opportunities.
- **Need for Solution:** Create a web-based solution to maintain alumni profiles, enable structured communication, and support mentorship and career guidance initiatives.

Group 40 & 8 – Campus Facility Booking System

- **Problem:** Booking labs, halls, or sports facilities manually leads to scheduling conflicts.
- **Need for Solution:** Implement a web-based solution to manage bookings with conflict detection, approvals, and visibility for all stakeholders to optimize resource usage

Group 25 & 10 – Student Scholarship/Financial Aid Application Tracker

- **Problem:** Students struggle to track the status of scholarship or aid applications.
- **Need for Solution:** Develop a web-based solution to submit applications, monitor approvals, and notify students of updates, improving transparency and accessibility.

Group 27 & 12 – Lost & Found Management Portal

- **Problem:** Lost items on campus are often unreported or mishandled.
- **Need for Solution:** Create a web-based solution to log, track, and claim lost or found items efficiently, minimizing lost property and improving accountability.

Group 13 & 30 – Student Group/Club Management System

- **Problem:** Clubs and student groups lack structured ways to manage members and activities.
- **Need for Solution:** Build a web-based solution to manage clubs, membership, events, and announcements efficiently, promoting engagement and organization.

Group 15 & 22 – Internship & Placement (WIL) Management System

- **Problem:** Students cannot easily track internship applications, company responses, or placement opportunities.
- **Need for Solution:** Develop a web-based solution to manage applications, approvals, deadlines, and progress, supporting student placement efficiently.

Group 34 & 18 – Mentorship & Peer Support Solution

- **Problem:** Students lack structured access to mentors or peer tutors.
- **Need for Solution:** Build a web-based solution to connect students with mentors, schedule sessions, track outcomes, and provide feedback.

Group 19 & 3 – Campus Security Alert System

- **Problem:** Immediate safety alerts are not effectively communicated to students.
- **Need for Solution:** Implement a web-based solution for real-time alerting of campus security incidents, allowing students and staff to respond quickly and improving overall safety.

Group 21 & 16 – Student Mental Wellness Tracker

- **Problem:** Students' mental health concerns are often addressed reactively; there is no structured tracking system.
- **Need for Solution:** Create a web-based solution to log wellness, schedule counselling sessions, and provide dashboards for monitoring trends and intervention needs.

Group 23 & 1 – Peer-Led Workshop Scheduling Solution

- **Problem:** Student-led workshops or skill-sharing sessions are not formally organized or tracked.
- **Need for Solution:** Build a web-based solution for students to create, schedule, and join workshops, including attendance tracking and feedback collection.

Group 9 & 26 – Campus Carpool & Transportation Solution

- **Problem:** Students and Staff face transportation challenges and lack a system for sustainable commuting.
- **Need for Solution:** Develop a web-based solution for ride-sharing, carpool scheduling, and campus transport notifications.

Group 29 & 14 – Faculty-Student Collaborative Project Hub

- **Problem:** Faculty-student collaboration opportunities are informal and poorly documented.
- **Need for Solution:** Create a web-based solution to post projects, allow students and staff to join, update milestones, and track outcomes collaboratively.

Group 31 & 5 – Student Volunteering & Community Service Solution

- **Problem:** Participation in volunteer programs is scattered and hard to track for recognition.
- **Need for Solution:** Develop a web-based solution to log volunteering hours, register for events, and generate certificates of participation.

Group 33 & 17 – Skill-Building Challenge Solution

- **Problem:** Students lack structured, gamified opportunities to build skills.
- **Need for Solution:** Create a web-based solution to post weekly challenges, track completion, and award badges or certificates.

Group 35 & 36 – Student Housing & Roommate Matching Solution

- **Problem:** Students struggle to find compatible roommates or suitable housing.
- **Need for Solution:** Build a web-based solution to match students based on preferences, track room allocation, and manage agreements.

Group 37 & 38 – Sustainable Campus Projects Tracker

- **Problem:** Sustainability initiatives lack visibility and tracking.
- **Need for Solution:** Develop a web-based solution to submit, monitor, and report sustainable projects across campus.

Group 39 & 7 – Community Skill Exchange Solution

- **Problem:** Local residents cannot easily share or request skills.
- **Need for Solution:** Build a web-based solution to post skills, request help, and track completed collaborations.

Group 41 & 11 – Local Waste Recycling Initiative Tracker

- **Problem:** Recycling participation is low due to lack of tracking and rewards.
- **Need for Solution:** Create a web-based solution to log recycling contributions, track points, and provide recognition.

Group 42 & 28 – Neighborhood Safety & Incident Reporting and Community Disaster Alert Solution

- **Problem:** Local safety issues and emergencies are reported inconsistently, slowing response.
- **Need for Solution:** Develop a web-based solution to log incidents, notify authorities, track resolutions, and provide real-time disaster alert

Project Objective

The objective of this project is to design and develop a web-based solution that addresses the assigned challenge. Each group is required to analyse the allocated problem, explore viable solution approaches, and implement a functional application that effectively responds to the identified needs.

Methodology

- The project must follow the Crystal (Yellow) Agile Development Methodology, promoting iterative development, continuous feedback, adaptability, and effective team collaboration.

Technology Stack

The web-based solution must be developed using the following core technologies and standards:

Core Development Technologies

- HTML5, CSS3, and JavaScript must be used to create a responsive and accessible user interface.
- UI frameworks such as Bootstrap may be used for consistency and responsiveness.
- Client-side validation should be implemented to enhance user experience and reduce invalid submissions.
- **Programming Language:** Python
- **Framework:** Flask (for backend development, routing, request handling, and integration of application components)
- A relational database management system (RDBMS) must be selected and implemented according to the system's data storage, integrity, scalability, and performance requirements (e.g., MySQL, PostgreSQL, or SQLite). Proper database design principles, including normalization and relationship modelling, must be applied.

Backend and API Requirements

The application may implement RESTful APIs to facilitate structured communication between the frontend and backend components. Endpoints should follow proper HTTP methods (GET, POST, PUT, DELETE) and return appropriate status codes.

External API Integration (where applicable)

The solution may integrate third-party APIs to enhance functionality, such as:

- Email notification services (e.g., automated status updates)
- SMS gateways
- Authentication services (e.g., OAuth-based login)
- Mapping services (for location-based features)
- Payment gateways (if relevant to the assigned challenge)

Authentication and Authorisation:

The system must implement secure user authentication and role-based access control (e.g., admin, user,).

Data Management and Security

- Where the application collects, stores, or processes personal information (e.g., names, student numbers, contact details, wellness records, complaints, applications), it must comply with the principles of the Protection of Personal Information Act (POPIA).
- Proper database schema design with primary and foreign keys.
- Secure password storage using hashing mechanisms (e.g., bcrypt).
- Input validation and sanitisation to prevent SQL injection and cross-site scripting (XSS).
- Session management and protection against common web vulnerabilities.

Development and Deployment Tools

- Git must be used for source code management and collaboration.
- Basic unit testing and functional testing should be conducted before deployment.
- The application should be deployable a cloud-based hosting service (e.g., PythonAnywhere, Render, or similar platforms).

System Design Expectations

- Modular code structure following best development practices.
- Separation of concerns (e.g., models, routes, templates).
- Clear file organisation and maintainable architecture

Groups are encouraged to apply analytical thinking, technical expertise, and creativity to produce solutions that contribute meaningfully to the DUT community and, where applicable, to broader societal contexts. Additional guidance and supporting resources may be provided where necessary.

Project proposal

To ensure timely completion, the proposed solution must be developed within the allocated timeframe. The group leader is responsible for formally communicating, via email, the proposed web-based application solution to the lecturer for review and preliminary approval. Feedback regarding approval or required revisions will be communicated within 24 hours.

The project proposal for the web-based application must briefly include the following components:

1. Purpose

A concise statement clearly defining the problem the web-based application aims to address, outlining its impact and justifying the need for an effective solution.

2. Solution Overview

A concise but comprehensive description of the proposed web-based solution, clearly explaining how the application will function and how it will address the identified problem effectively

3. Key Benefits and Functionality

An overview of the primary advantages and distinguishing features of the application, emphasising its contribution to improved efficiency, enhanced user experience, and strengthened problem-solving capability.

4. Milestones

A structured development timeline highlighting key phases, including research, requirements analysis, design, development, testing, deployment, and evaluation.

5. Mock-ups or Diagrams

Clear and relevant visual representations (e.g., wireframes, system architecture diagrams, database schemas, or user flow diagrams) that illustrate the system's structure, functionality, and user interaction logically and professionally

The deadline for the approval of potential solutions: 2 March 2026. Submissions must be in “pdf” format.

Further details regarding the final documentation requirements and presentation guidelines will be communicated in due course